1. Overview of the Program

The Student Grade Tracker is a command-line application designed to help students and instructors manage academic assignments, calculate cumulative percentage grades, and determine final letter grades. The program utilizes object-oriented programming principles in C++, encapsulating assignment data and grading logic within clean, modular classes.

2. Key Programming Concepts Used

- **Classes and Objects:** Encapsulation of student data and assignment management behaviors within a dedicated `Student` class.
- **Structures (`struct`):** Use of custom data structures to group related assignment attributes including name, earned score, and maximum possible score.
- **Vectors:** Dynamic arrays (`std::vector`) used to store and manage a variable number of assignments efficiently.
- **Sorting and Binary Search:** Alphabetical sorting of assignment records followed by an implementation of binary search to look up specific assignments quickly by name.
- **Formatting Output:** Utilization of `std::fixed`, `std::setprecision`, and `std::setw` for clean, aligned console reports.

3. Screenshot of Program

